

Space surveillance telescope

DARPA's new light sensitive telescope built for the US Air force can observe faint objects in deep space which were impossible to observe

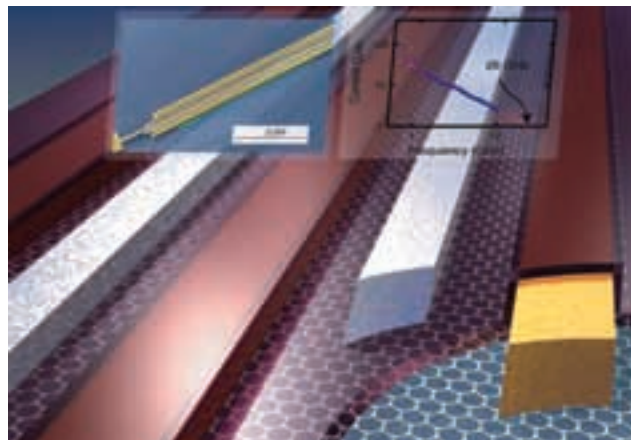
One ID for your web presence

US renews call for single online ID for its citizens to enable safe online presence and remove the need for multiple ID's

a technology for on-chip integration of ultra-compact nanophotonic circuits for manipulating the light signals, similar to the way electrical signals are manipulated in computer chips. According to IBM they are "developing nanoscale silicon photonics circuits to enable the integration of complete optical systems on a monolithic semiconductor chip that would eventually allow to overcome severe constraints of today's mostly copper I/O interconnects".

The technology is aimed at producing an exascale supercomputer which will be capable of performing one million trillion calculations in one second. Million trillion, though sounding like a fictional figure like a gazillion, is an actual unit of measure.

It's not hard to imagine this technology coming into the hands of consumers in a while. Chips based on this technology will have three layers. The first will be made up of the cores (multiple cores for supercomputers) and on top of which will be a memory layer for easier access to the cache. On top of the stack is the Photonic layer with many thousands of individual optical devices (modulators, detectors, switches) as well as analogue electrical circuits (amplifiers, drivers, latches, etc.). This layer will route data traffic both on and off the chip.



Graphene Transistor

Move everything closer

As we mentioned earlier on, apart from processors, memory and data storage constitute the other side of the computing chain. As things work today, especially at the data centre-level computers, data is shuttled between where it is stored to where it is processed at great real and notional costs viz. energy and time. Processors cache frequently used information close-by (volatile) but once processed, it is moved to a more permanent storage to be called upon when required.

HP has some interesting concepts brewing on this end. Parthasarathy Ranganathan, a researcher from HP is working on a different architectural

approach to merge memory storage and processing.

"We are entering an exciting era for systems design. In addition to continued advances in performance, next-generation designs are also addressing important challenges around power, sustainability, manageability, reliability, and scalability. At the same time, a confluence of emerging tech-

nologies (such as photonics, non-volatile storage and 3D stacking), and new workloads (around cloud computing, unstructured data, and virtualization) offer additional new opportunities and challenges. The confluence of these trends motivates a rethinking of system design -- motivated by holistic considerations and cross-cutting traditional design boundaries", said Ranganathan.

In a white paper published in IEEE CS, he has described the concept of using nanostores. "The key property of nanostores is the colocation of processors with nonvolatile storage, eliminating many intervening levels of the storage hierarchy. All data is stored in a single-level non-volatile memory datastore that replaces traditional disk and DRAM layers -- disk use is relegated to archival backups", he says. The nanostore is based on a nanoscale technology called memristor - a device that can perform logic operations as well as store data. What would this mean to systems? "Our proposed new design, nanostores, have the potential for dramatic (10-100X) improvements in efficiency and present significant opportunities for future research", Partha informed us. Given a few assumptions HP expects in the future a node that groups nine nanostores, can provide one-half to one terabyte of non-volatile memory with teraflops of local compute co-located with the storage and about 256 Gbytes of aggregate datastore bandwidth. Indeed a bountiful future to look forward to. **[1]**



Silicon optical transceivers will have modulators, switches and detectors all cast the same CMOS die